

TEMASEK JUNIOR COLLEGE
2022 JC2 PRELIMINARY EXAMINATION
Higher 2



Chemistry

Paper 1 Multiple Choice

9729/01

16 September 2022

1 hour

Additional Materials: Multiple Choice Answer Sheet (OMS)
Data Booklet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, glue or correction fluid.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C, D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate OMS.

Read the instructions on the OMS very carefully.

Write your name & Civics Group on the OMS. Shade your index number in the appropriate boxes.

1. Enter your NAME (as in NRIC). _____
2. Enter the SUBJECT TITLE. _____
3. Enter the TEST NAME. _____
4. Enter the CLASS. _____

Write your **name**
and **Civics Group**

5. Enter your CLASS NUMBER or INDEX NUMBER.

6. Now SHADE the corresponding lozenge in the grid for EACH DIGIT or LETTER

WRITE		SHADE APPROPRIATE BOXES									
I N D E X	0	1	2	3	4	5	6	7	8	9	
	0	1	2	3	4	5	6	7	8	9	
	0	1	2	3	4	5	6	7	8	9	
N U M B E R	0	1	2	3	4	5	6	7	8	9	
	0	1	2	3	4	5	6	7	8	9	
	A	B	C	D	E	F	G	H	I		

Index number
(refer to entry proof)

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

The use of an approved scientific calculator is expected, where appropriate.

This document consists of **16** printed pages.

Answer Key

1	2	3	4	5		6	7	8	9	10
D	C	A	B	D		B	B	C	A	A
11	12	13	14	15		16	17	18	19	20
D	C	B	D	B		A	C	A	B	D
21	22	23	24	25		26	27	28	29	30
C	B	D	C	A		D	C	B	C	A

1 Use of the Data Booklet is relevant to this question.

Atoms or ions that contain the same number of neutrons are isotonic.

Which of the following pairs of ions fulfil both criteria below?

- The two ions are isotonic.
- The first ion deflects more than the second ion when passed through an electric field.

A $^{35}\text{Cl}^-$, $^{31}\text{P}^-$

B $^{31}\text{P}^-$, $^{31}\text{P}^{3-}$

C $^{31}\text{P}^{3-}$, $^{32}\text{P}^{2-}$

D $^{31}\text{P}^{3-}$, $^{32}\text{S}^{2-}$

Answer: D

Worked Solution:

Number of neutrons in $^{31}\text{P}^{3-} = 31 - 15 = 16$

Number of neutrons in $^{32}\text{S}^{2-} = 32 - 16 = 16$

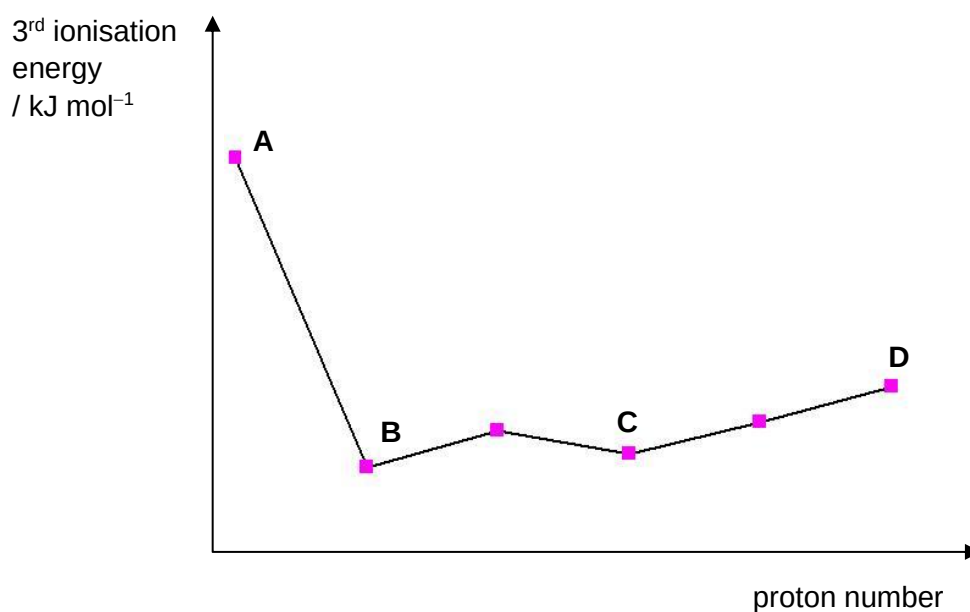
Thus, $^{31}\text{P}^{3-}$ and $^{32}\text{S}^{2-}$ are isotonic.

Angle of deflection of species in electric field $\propto \frac{\text{charge}}{\text{mass}}$

$^{31}\text{P}^{3-}$ has a higher charge and a smaller mass than $^{32}\text{S}^{2-}$. The angle of deflection of $^{31}\text{P}^{3-}$ will be greater than that for $^{32}\text{S}^{2-}$.

- 2 The variation in the third ionisation energy of six consecutive elements in the Periodic Table is shown in the graph.

Which of these elements is a Group 15 element?



Answer: C

Worked Solution: 3rd electron of B comes from an inner shell, B has 3 valence electron and is from Group 13. C is from Group 15.

- 3 Which group of particles is in order of increasing first ionisation energy?

1 Ga Ge As

2 As Se Br

3 Se Br Kr

4 P As Sb

A 1 and 3

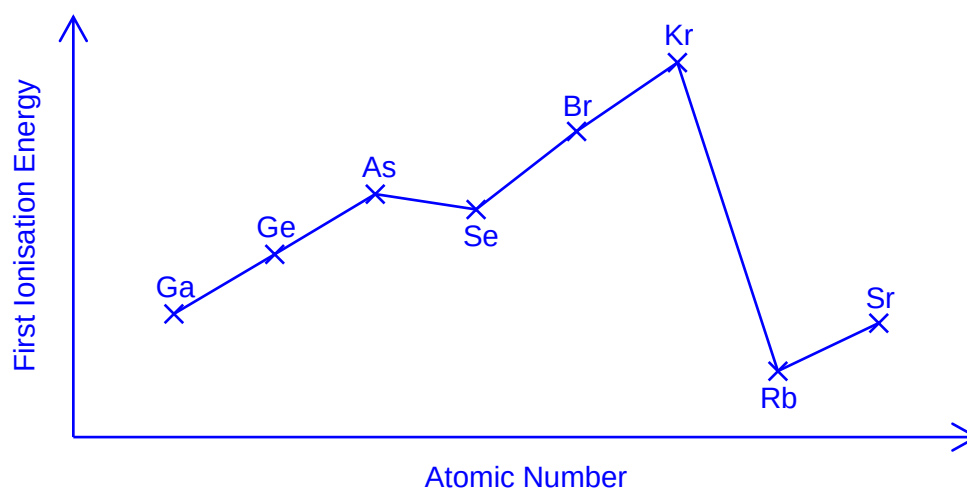
B 2 and 4

C 2 and 3

D 1 and 4

Answer: A

Worked Solution:



1: correct, from diagram, IE of Ga < Ge < As

2: wrong, Se < As < Br

3: correct, Se < Br < Kr

4: wrong, Sb < As < P

- 4 Carbon forms double bonds with oxygen, and with two other Group 16 elements, **Y** and **Z**. In the molecules CO₂, CO**Y** and CO**Z**, the polarities of these double bonds do not necessarily cancel.

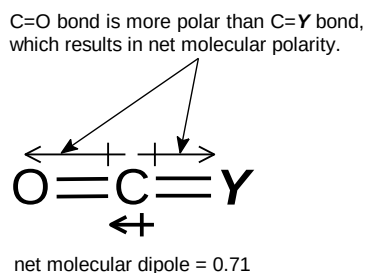
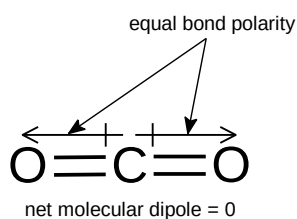
	overall polarity of molecule
CO ₂	0
CO Y	0.71
CO Z	0.73

Which factor best accounts for these observations?

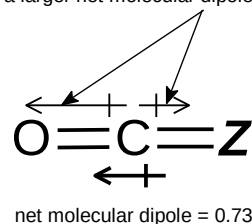
- A The C=O bond is the least polar, while the C=**Z** bond is the most polar.
- B Electronegativity increases in the order: **Z** < **Y** < O
- C The C=O, C=**Y** and C=**Z** bonds are all polar.
- D Bond length increases in the order: C=O < C=**Y** < C=**Z**

Answer: **B**

[Worked Solution:](#)



C=O bond is more polar than C=Z bond, which results in net molecular polarity. However, C=Z bond is less polar than C=Y bond, which results in a larger net molecular dipole.



✗	A	O is the most electronegative element in Group 16. Thus, the C=O bond is the most polar amongst the three bonds.
✓	B	O is the most electronegative element in Group 16. Since C=Z bond is less polar than C=Y bond, this suggests that Y is more electronegative than Z. (See diagrams above.)
✗	C	While this statement is true, it does not explain the observed trend.
✗	D	Bond length has little bearing on net molecular dipole.

5 Which of the following best describes the change in bond angle when SO₂ is converted to SO₃?

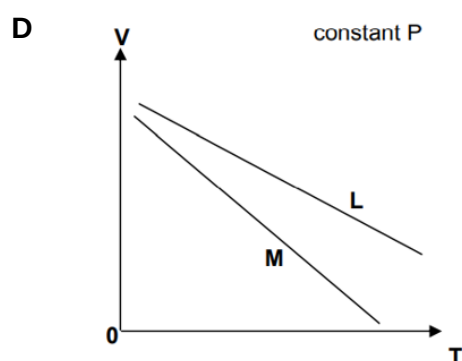
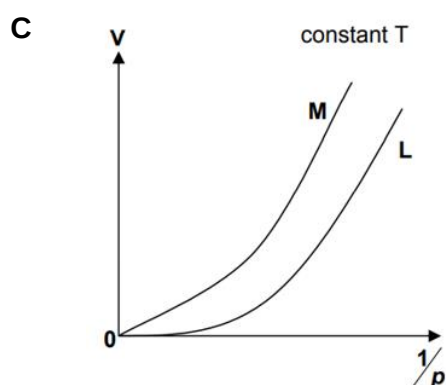
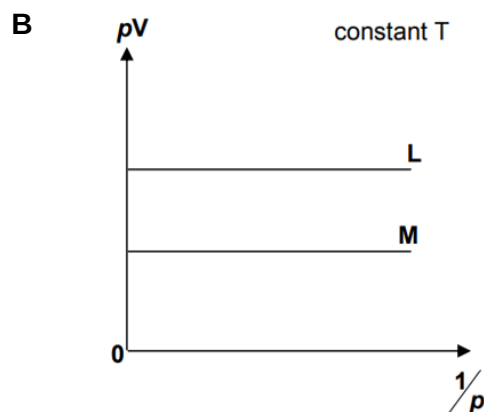
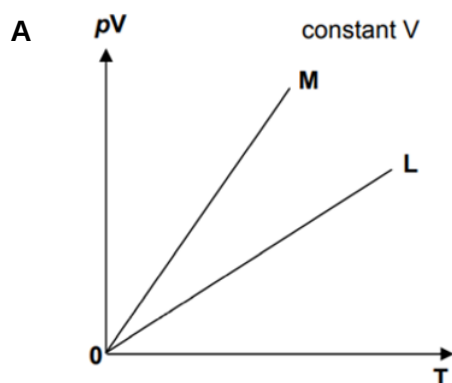
- A Decreases to 107°
- B Increases to 107°
- C Decreases to 120°
- D Increases to 120°

Answer: **D**

Worked Solution:

SO₂ has bent shape, bond angle < 120°. SO₃ has trigonal planar shape, bond angle is 120°.

- 6 Which graph correctly describes the behaviour of fixed masses of the ideal gases **L** and **M** where the number of moles of **L** is greater than number of moles of **M**?



Answer: **B**

Worked Solution:

$$PV = nRT$$

- A. Incorrect. The number of moles of **L** is greater than number of moles of **M**
- B. Correct. At constant T , $PV = n \times \text{constant}$
- C. Incorrect. At constant T , $V = nRT/P$, straight line pass through origin, different gradient for **L** & **M**
- D. Incorrect. At constant P , $V = (nR)/p \times T = \text{constant} \times T$

- 7 **X**, **Y** and **Z** are elements in Period 3 of the Periodic Table. The table below shows the changes in pH when compounds of these elements are added to water.

element	change in pH of solution when oxide of element is added to water	change in pH of solution when chloride of element is added to water
X	increases	no change
Y	decreases	decreases
Z	no change	decreases

What could be the identities of **X**, **Y** and **Z**?

	X	Y	Z
A	Na	Al	Mg
B	Na	P	Al
C	Si	P	Al
D	P	Na	Si

Answer: **B**

Worked Solution:

element		pH of solution when oxide of element is added to water	pH of solution when chloride of element is added to water
X	Na	$\text{Na}_2\text{O} + \text{H}_2\text{O} \rightarrow 2\text{NaOH}$ pH = 13	NaCl(aq) , pH = 7
Y	P	$\text{P}_4\text{O}_{10} + 6\text{H}_2\text{O} \rightarrow 4\text{H}_3\text{PO}_4$ pH = 2	$\text{PCl}_5 + 4\text{H}_2\text{O} \rightarrow \text{H}_3\text{PO}_4 + 5\text{HCl}$ pH = 2
Z	Al	Al_2O_3 is insoluble in water pH = 7	AlCl_3 undergoes hydration and hydrolysis. pH of solution is about 3.

8 Which of the following statements about Group 2 elements is correct?

- A Solubility of hydroxides decreases down the group.
- B Covalent character of chlorides increases down the group.
- C Thermal stability of carbonates increases down the group.
- D Electrode potential, $E^\ominus(\text{M}^{2+}/\text{M})$, becomes more positive down the group.

Answer: **C**

Worked Solution: Ionic radii of group 2 cations increases down the group. Charge density and polarising power decrease down the group, thermal stability increases down the group.

- 9 Which of the following statements about the Group 17 elements and compounds is **incorrect**?
- A** When aqueous sodium chloride is added to aqueous bromine, the orange solution is decolourised.
- B** When aqueous silver nitrate is added to aqueous sodium chloride followed by nitric acid, a white precipitate remains.
- C** When aqueous chlorine is added to excess sodium thiosulfate, followed by barium nitrate, a white precipitate is formed.
- D** When chlorine gas is bubbled through potassium iodide, the solution turns brown.

Answer: **A**

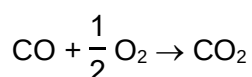
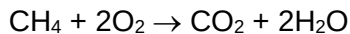
A. Incorrect statement - Chlorine has a greater reducing strength than bromine. Bromine will not be able to displace chloride.

B. AgCl is insoluble in nitric acid.

C. Chlorine oxidises thiosulfate to sulfate, which forms white ppt of barium sulfate.

D. Chlorine displaces iodide, brown iodine solution is formed.

- 10 A 15 cm^3 mixture of carbon monoxide and methane was mixed with excess oxygen and exploded.



There was a contraction in volume of 15 cm^3 at room temperature and pressure after the reaction.

What is the percentage by volume of methane in the mixture?

- A** 33 % **B** 50 % **C** 60 % **D** 75 %

Answer: **A**

Worked Solution:

Let $V_{\text{CH}_4} \text{ cm}^3$ be the volume of CH_4 in mixture.

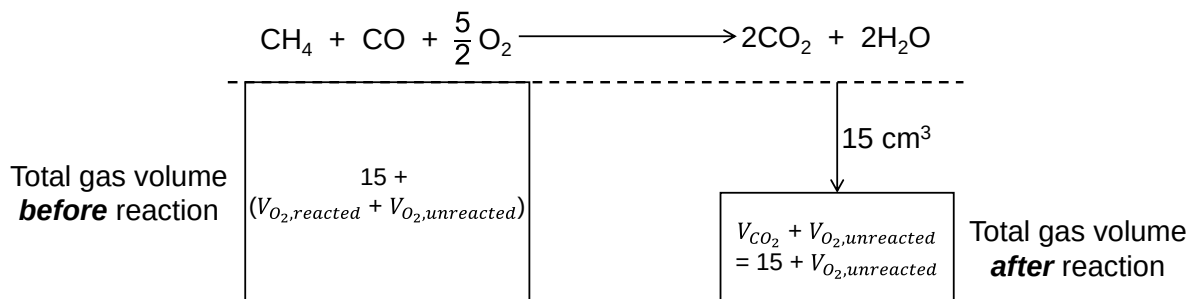
$\Rightarrow$ Volume of CO in mixture = $(15 - V_{\text{CH}_4}) \text{ cm}^3$

	CH_4	:	O_2	:	CO_2
mole ratio	1	:	2	:	1
volume ratio	V_{CH_4}	:	$2 V_{\text{CH}_4}$	:	V_{CH_4}

	CO	:	O_2	:	CO_2
mole ratio	1	:	$\frac{1}{2}$	:	1
volume ratio	$15 - V_{\text{CH}_4}$	:	$\frac{1}{2} (15 - V_{\text{CH}_4})$	:	$15 - V_{\text{CH}_4}$

Since total volume of CH_4 and CO combusted = 15 cm^3 , and equimolar ratios of CH_4 to CO_2 , and of CO to CO_2 are formed, total volume of CO_2 formed = 15 cm^3

Total volume of O_2 added into reaction vessel = $(V_{O_2,reacted} + V_{O_2,unreacted}) \text{ cm}^3$



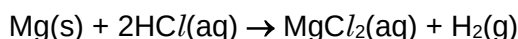
$$15 + V_{O_2,reacted} + V_{O_2,unreacted} = 15 + (15 + V_{O_2,unreacted})$$

$$\Rightarrow V_{O_2,reacted} = 15 \text{ cm}^3$$

$$\Rightarrow 2 V_{\text{CH}_4} + \frac{1}{2} (15 - V_{\text{CH}_4}) = 15$$

$$\Rightarrow V_{\text{CH}_4} = 5 \text{ cm}^3, \text{ which is } \frac{5}{15} \times 100 = \underline{\underline{33.3 \%}} \text{ of the mixture}$$

- 11 Magnesium reacts readily with dilute hydrochloric acid to form magnesium chloride and hydrogen.



The standard enthalpy change for this reaction can be measured in the laboratory.

What further information is needed in order to calculate the standard enthalpy change of formation of magnesium chloride?

- 1 ΔH_f for HCl(aq)
- 2 lattice energy of MgCl_2
- 3 enthalpy change of solution of MgCl_2

- A 1, 2 and 3 B 1 and 2 only C 2 and 3 only D 1 only

Answer: **D**

Worked Solution:

$$\Delta H_r^\ominus = \sum m \Delta H_f^\ominus(\text{pds}) - \sum n \Delta H_f^\ominus(\text{rxts})$$

$$\Delta H_f^\ominus(\text{MgCl}_2) = \Delta H_r^\ominus + \Delta H_f^\ominus(\text{HCl})$$

The standard enthalpy change of formation of elements in their standard states at 298 K and 1 bar is, by definition, zero

12



The mechanism for the above reaction involves the following steps.



Based on the information, what is the rate equation for this reaction?

A rate = $k [\text{A}]^2 [\text{B}]^2$

B rate = $k [\text{B}] [\text{C}]$

C rate = $k [\text{A}]^2 [\text{B}]$

D rate = $k [\text{A}] [\text{B}]^2$

Answer: **C**

Worked Solution:

The rate of the reaction depends on the rate of the slow step, i.e.

$$\text{rate} = k' [\text{B}] [\text{C}]$$

However, the rate equation for the overall reaction should be in terms of reactant concentrations only. (Note that **C** and **D** are reaction intermediates.)

Since step 1 of the mechanism is an equilibrium, $2\text{A} \rightleftharpoons \text{C}$, we can write an expression for its equilibrium constant,

$$K_{\text{step 1}} = \frac{[\text{C}]}{[\text{A}]^2}$$

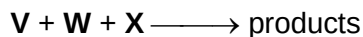
$$\text{Rearranging, } [\text{C}] = K_{\text{step 1}} [\text{A}]^2$$

The rate equation then becomes

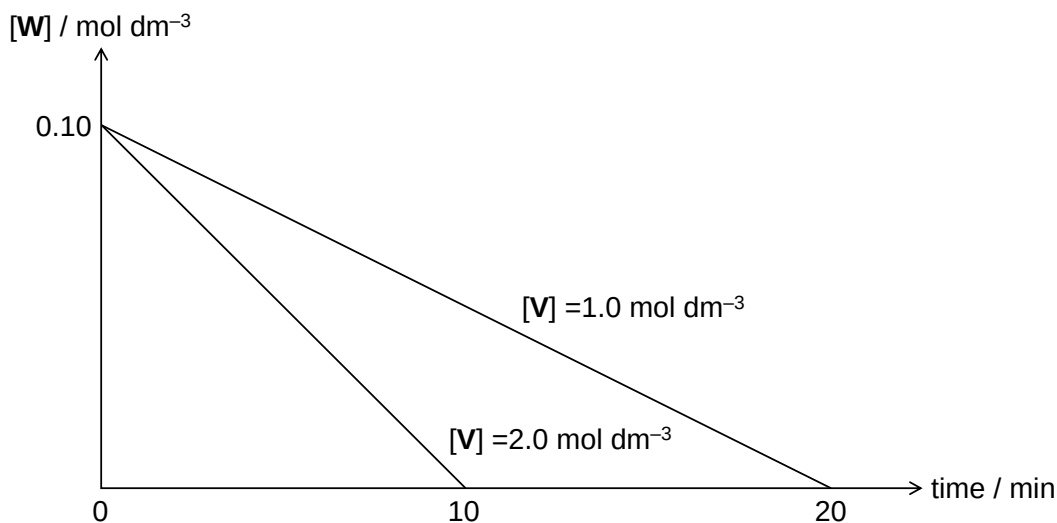
$$\text{Rate} = k' [\text{B}] (K_{\text{step 1}} [\text{A}]^2) = k [\text{A}]^2 [\text{B}]$$

$$\text{where } k = k' \times K_{\text{step 1}}$$

13 Consider the reaction



Two separate experiments were carried out using 1.0 mol dm^{-3} and 2.0 mol dm^{-3} of $\mathbf{V}$ respectively. A graph of the variation of the concentration of $\mathbf{W}$ with time is plotted.



Which of the following statements can be determined from the above information?

- 1 The reaction is first order with respect to $\mathbf{V}$.
- 2 The reaction is zero order with respect to $\mathbf{W}$.
- 3 The reaction is first order with respect to $\mathbf{X}$.

- A 1, 2 and 3
 B 1 and 2 only
 C 2 and 3 only
 D 1 only

Answer: **B**

Worked Solution:

Graph of $[\mathbf{W}]$ against time is a downward-sloping straight line with constant gradient.

⇒ Rate of reaction is independent of $[\mathbf{W}]$.

⇒ Reaction is zero order wrt $[\mathbf{W}]$.

$[\mathbf{W}] / \text{mol dm}^{-3}$	$[\mathbf{V}] / \text{mol dm}^{-3}$	Rate of reaction / $\text{mol dm}^{-3} \text{ min}^{-1}$
0.10	1.0	$\frac{0.10}{20} = 0.0050$
0.10	2.0	$\frac{0.10}{10} = 0.010$

When $[V] \times 2$, while $[W]$ is unchanged, rate of reaction $\times 2$

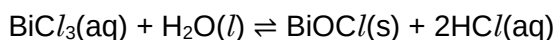
$\Rightarrow$ Reaction is first order wrt $[V]$.

Order of reaction wrt $[X]$ cannot be determined from the graph.

$\therefore$ Only Statements 1 & 2 are correct.

- 14** When 0.1 mol of bismuth chloride is added to 2 dm³ of water, it reacts to form 0.02 mol of white precipitate of bismuth oxychloride and a solution of hydrochloric acid.

The equation for the reaction is as follows:



What is the correct expression for the equilibrium constant K_c ?

A $\frac{(2 \times 0.02)^2}{0.08}$

B $\frac{(0.02)(2 \times 0.02)^2}{0.08}$

C $\frac{\left(\frac{0.02}{2}\right)\left(\frac{2 \times 0.02}{2}\right)^2}{\frac{0.08}{2}}$

D $\frac{\left(\frac{2 \times 0.02}{2}\right)^2}{\frac{0.08}{2}}$

Answer: **D**

Worked Solution:

	$\text{BiCl}_3(\text{aq})$	+	$\text{H}_2\text{O}(\text{l})$	$\rightleftharpoons$	$\text{BiOCl}(\text{s})$	+	$2\text{HCl}(\text{aq})$
Initial amt / mol	0.1		–		–		0
Change in amt / mol	–0.02		–		+0.02		+2(0.02)
Eqm amt / mol	0.08		–		0.02		2(0.02)

$$K_c = \frac{[\text{HCl}]^2}{[\text{BiCl}_3]} = \frac{\left(\frac{2 \times 0.02}{2}\right)^2}{\frac{0.08}{2}}$$

- 15 The enthalpy change of dissociation of ethanoic acid in water is $+1.0 \text{ kJ mol}^{-1}$.

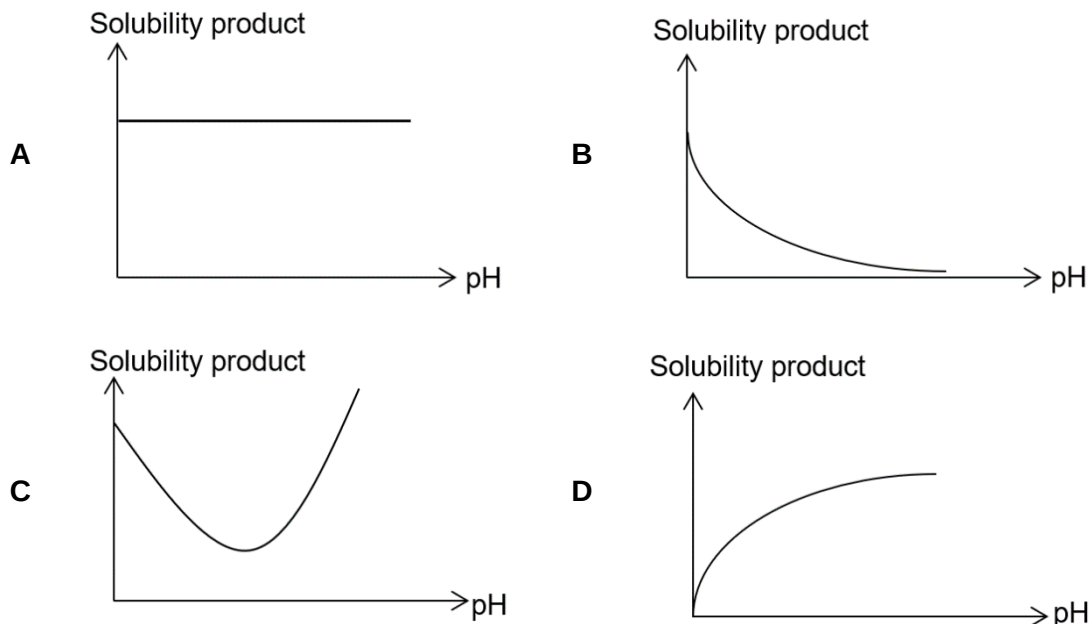
Which correctly describes the value of acid dissociation constant and degree of dissociation when temperature increases?

	acid dissociation constant	degree of dissociation
A	increase	remain constant
B	increase	increase
C	decrease	remain constant
D	decrease	increase

Answer: **B**

Worked Solution: When temp increases, by LCP, position of eqm shifts to the right to favour the endothermic reaction to absorb heat. So K_a increases. Since $K_a = \alpha^2 c$, degree of dissociation also increases.

- 16 The numerical value of the solubility product of copper(I) iodide is 1.27×10^{-12} . Given that HI is a strong acid, which diagram shows how the solubility product of CuI will vary with pH at constant temperature?



Answer: **A**

Worked Solution:

Solubility product is only affected by changes in temperature.

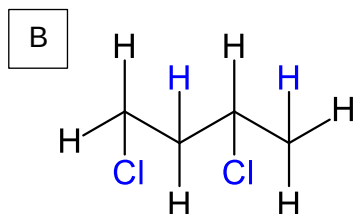
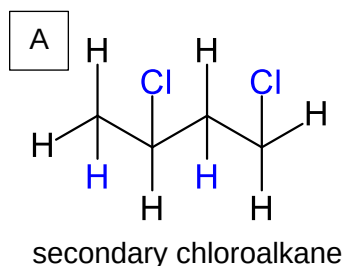
- 17 Buta-1,3-diene, $\text{CH}_2=\text{CH}-\text{CH}=\text{CH}_2$, reacts completely with two moles of hydrogen chloride. Which of the following about the products formed is correct?

	No. of constitutional isomers	No. of products that are secondary chloroalkanes
A	2	1
B	3	1
C	3	2
D	4	2

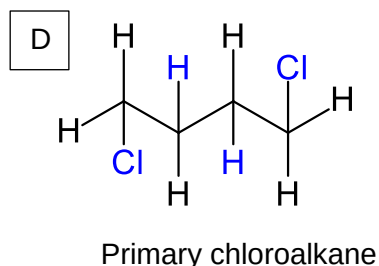
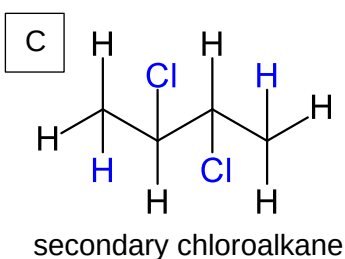
Answer: **C**

[Worked Solution:](#)

Products formed:



Isomer A and B are the same.
No. of constitutional isomers: 3



- 18 Both cyclohexene and benzene react with bromine, but under different conditions. Cyclohexene reacts readily with liquid bromine while benzene requires a catalyst before a reaction with bromine can occur. Which of the following explains why cyclohexene is more reactive than benzene?
- A** The localised π electron cloud in cyclohexene makes the molecule more susceptible to attack by electrophiles.
 - B** Cyclohexene is not a planar molecule while all atoms in benzene lie on the same plane.
 - C** The delocalised π electron cloud in benzene repels electrophiles.
 - D** Cyclohexene has a higher boiling point than benzene.

Answer: **A**

Worked Solution:

Due to the delocalisation of π electrons, the π electron cloud in benzene is less susceptible to an electrophilic attack than a localised π electron cloud in cyclohexene.

Thus, cyclohexene reacts readily with a weak electrophile like a polarised Br_2 , while benzene requires a Lewis acid catalyst to generate a stronger electrophile, Br^+ , before a reaction with bromine can occur.

- 19** Chloroalkane **W**, $\text{C}_5\text{H}_{11}\text{Cl}$, rotates plane polarised light. On heating **W** with alcoholic NaOH , two alkenes **U** and **X** are formed, with **U** being formed in larger amount.

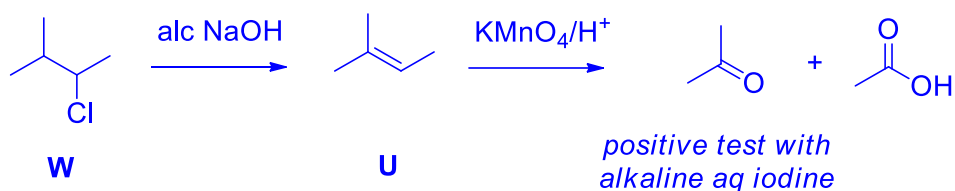
A yellow precipitate is produced when aqueous alkaline iodine is warmed with one of the products of the reaction between **U** and hot acidified KMnO_4 .

Which of the following is **W**?

- A** $\text{CH}_3\text{CH}_2\text{CHClCH}_2\text{CH}_2\text{CH}_3$
B $\text{CH}_3\text{CH}(\text{CH}_3)\text{CHClCH}_3$
C $\text{CH}_3\text{CH}_2\text{CHClCH}_2\text{CH}_3$
D $\text{CH}_3\text{CCl}(\text{CH}_3)\text{CH}_2\text{CH}_3$

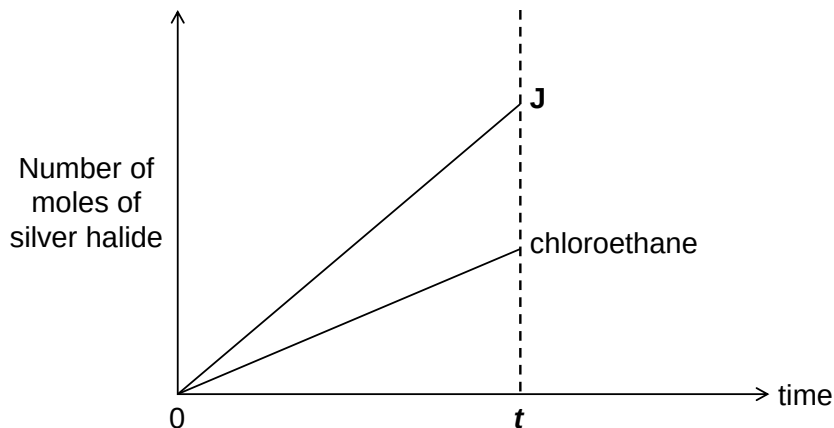
Answer: **B**

Worked Solution:



- 20 When halogenoalkanes are heated with ethanolic silver nitrate, insoluble silver halides are formed.

A student investigated the rate of formation of silver halides when the same amount of chloroethane and **J** were reacted separately with hot ethanolic silver nitrate. Each reaction was monitored till time, t , and the following graphs were obtained.



Which of the following compounds could be **J**?

- 1 $\text{C}/\text{CH}_2\text{CH}_2\text{C}/$
- 2 $\text{CH}_3\text{CH}_2\text{Br}$
- 3 $\text{CH}_3\text{CH}_2\text{I}$

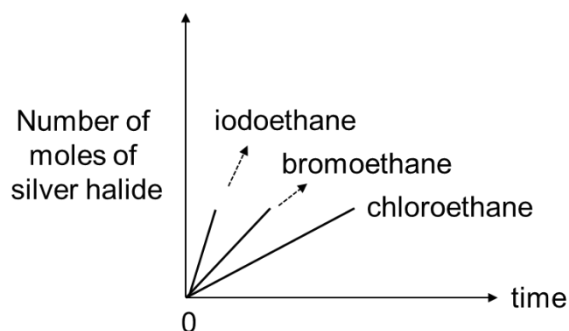
- A 1, 2 and 3
 B 1 and 2
 C 2 and 3
 D 1 only

Answer: **D**

Worked Solution:

Each mole of dichloroethane provides 2x the amount of AgCl as chloroethane, for the same amount of time of the experiment.

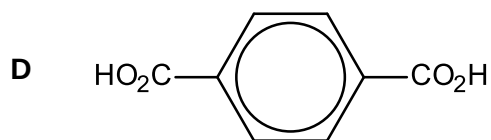
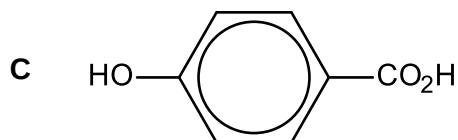
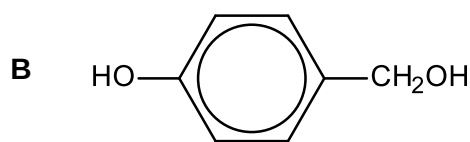
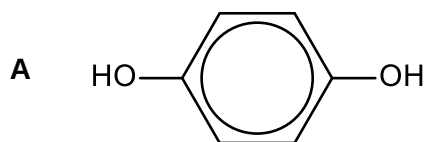
For bromoethane and iodoethane, the rate of reaction is faster due to weaker C–X bond, but the amount of AgX produced is the same as chloroethane, so the graph should be:



21 1 mole of each of the following four compounds is reacted separately with

- an excess of Na, and
- an excess of NaHCO_3 .

Which compound produces the same volume of gas with each of the two reagents?



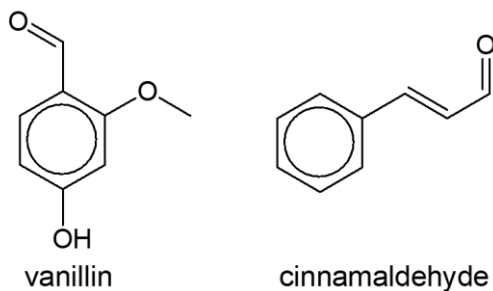
Answer: **C**

Worked Solution:

	$-\text{OH} \equiv \text{Na} \equiv \frac{1}{2}\text{H}_2$	$-\text{CO}_2\text{H} \equiv \text{NaHCO}_3 \equiv \text{CO}_2$
alcohol	√	X
phenol	√	X
acid	√	√

Option	H_2 from Na	CO_2 from NaHCO_3
A	1	0
B	1	0
C	1	1
D	1	2

- 22 Vanillin and cinnamaldehyde are some naturally occurring flavouring agents.



Which reagent, when added separately to the two compounds, would enable them to be distinguished from one another?

- A 2,4-dinitrophenylhydrazine
- B Fehling's solution
- C Hot acidified $K_2Cr_2O_7$
- D Tollens' reagent

Answer: **B**

Worked Solution:

- A × **2,4-DNPH:** positive test for carbonyl group in vanillin and cinnamaldehyde
- B ✓ **Fehling's solution:** positive test for aliphatic aldehyde group in cinnamaldehyde and negative for benzaldehyde in vanillin
- C × **Hot acidified $K_2Cr_2O_7$:** positive test for aldehyde group in vanillin and cinnamaldehyde.
- D × **Tollens' reagent:** positive test for both

23 What is the relative order of decreasing acidity for the following compounds?

- 1 2-nitrophenol
- 2 benzoic acid
- 3 2-chlorobenzoic acid
- 4 cyclohexanol

	most acidic	—————→		least acidic	
A	2	3	4	1	
B	2	3	1	4	
C	3	2	4	1	
D	3	2	1	4	

Answer: **D**

Worked Solution:

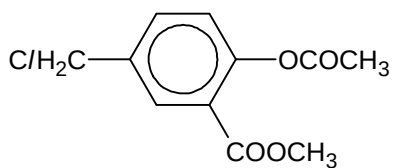
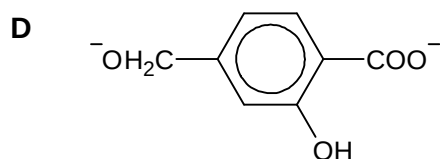
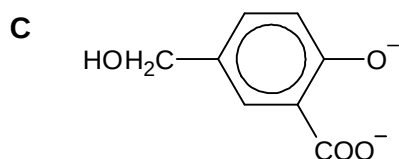
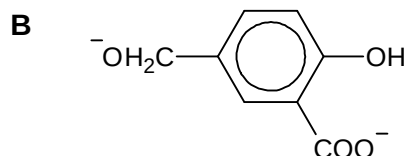
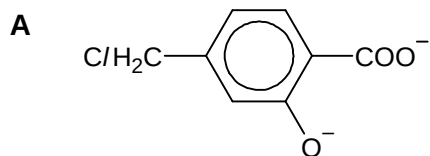
Order of decreasing acidity: carboxylic acids > phenols > alcohols

[Refer to page 5, Section 3.1 (Comparison of Acid strength between RCOOH, phenol and alcohols) in the Carboxylic Acids lecture notes.]

2-chlorobenzoic acid is more acidic than benzoic acid. 2-chlorobenzoic acid has an electron-withdrawing Cl atom which disperses the negative charge on the 2-chlorobenzoate ion and helps to stabilise it to a larger extent than the benzoate ion.

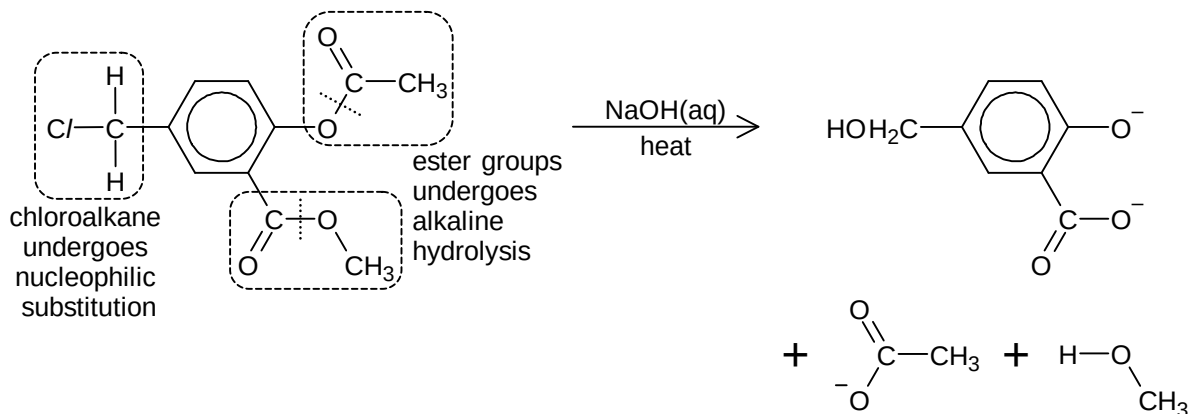
∴ Order of decreasing acidity: 3 > 2 > 1 > 4

- 24 Which one of the following represents the organic product when an excess of hot aqueous sodium hydroxide is added to compound **R**?

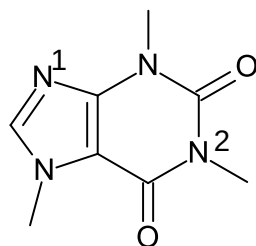
**R**

Answer: **C**

Worked Solution:



- 25 Caffeine is a central nervous system stimulant and is used as a cognitive enhancer to increase alertness. All the atoms in the 2 rings are planar.



caffeine

Which of the following statements about caffeine are correct?

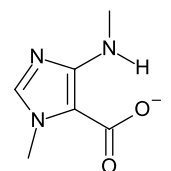
- 1 N^1 is more basic than N^2 .
 - 2 Caffeine undergoes a condensation reaction with ethanoyl chloride.
 - 3 1 mole of caffeine reacts with excess hot aqueous sodium hydroxide to form 2 moles of methylamine.
- A 1 only
- B 2 only
- C 1 and 3
- D 1, 2 and 3

Answer: **A**

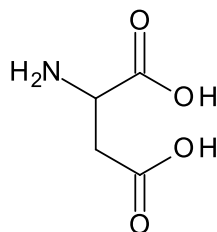
Worked Solution:

- 1 N^2 is the less basic as it is an amide where the lone pair of electrons on N is delocalised over the O–C–N bond and hence is not available for dative bonding with a proton.
 N^1 is the more basic as the lone pair of electrons on N is in an orbital that is perpendicular to the C=C and hence cannot be delocalised. The lone pair of electrons is available for dative bonding to H^+ .
- 2 Caffeine does not react with ethanoyl chloride as there are no H attached to N^1 and N^2 .

- 3 Caffeine undergo alkaline hydrolysis to form only one mole of methylamine and



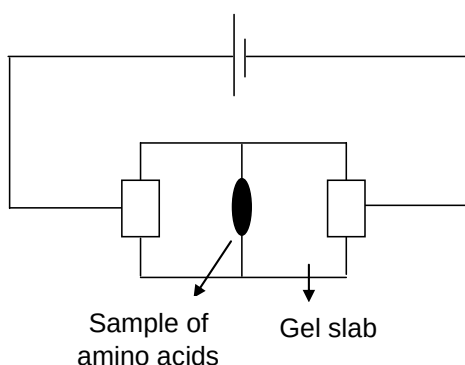
- 26 Aspartic acid is an α -amino acid that occur naturally in the body.



Aspartic acid

The three pK_a values associated with aspartic acid are 1.88, 3.65 and 9.60.

Electrophoresis is a technique of separating amino acids. Samples are placed on a gel slab containing a buffer and the ends of the gel slab are connected to charged electrodes. Positively charged species migrate towards the negative electrode while negatively charged species migrate towards the positive electrode. Ions of higher charge to mass ratio will migrate at a faster rate.



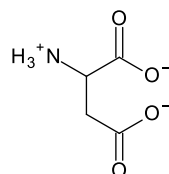
A solution of aspartic acid buffered at $pH = 4$ was placed in the electrophoresis set-up. Which of the following is correct?

- A The major species has a net charge of +1.
- B The major species present will migrate to the cathode.
- C The major species present will migrate to neither electrodes.
- D The major species will migrate faster if the pH was 11.

Answer: **D**

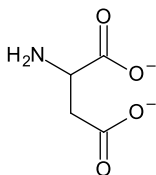
Worked Solution:

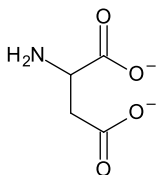
At $pH = 4$, the major species of aspartic acid present is



of -1 , it will migrate to the positive electrode (anode).

. Since it has a net charge



At pH = 11, the major species is . Since it is doubly charged, it would migrate faster than if the pH = 4.

27 Use of the Data Booklet is relevant to this question.

Which of the following is the strongest reducing agent?

- A hexacyanoferrate(III)
- B manganese(II)
- C nitrogen dioxide
- D vanadyl, VO^{2+}

Answer: C

Worked Solution:

Hexacyanoferrate(III) is an oxidising agent, NOT an reducing agent.

The stronger the reducing agent, the more likely it will undergo oxidation, the less positive/more negative the $E^\ominus$ will be.

$$E^\ominus(\text{NO}_3^-/\text{NO}_2) = +0.81 \text{ V} < E^\ominus(\text{VO}_2^+/\text{VO}^{2+}) = +1.00 \text{ V} < E^\ominus(\text{MnO}_2/\text{Mn}^{2+}) = +1.23 \text{ V}$$

28 Use of the Data Booklet is relevant to this question.

The reaction between iodide, I^- and peroxydisulfate, $\text{S}_2\text{O}_8^{2-}$ is very slow at room conditions. Which of the ions is unable to alter the rate of reaction between iodide and peroxydisulfate?

- A Co^{2+}
- B Cu^{2+}
- C Fe^{3+}
- D Fe^{2+}

Answer: B

Worked Solution:

Electrode half-equation	$E^\ominus / \text{V}$
$\text{Co}^{3+}(\text{aq}) + \text{e}^- \rightarrow \text{Co}^{2+}(\text{aq})$	+1.82
$\text{Cu}^{2+}(\text{aq}) + 2\text{e}^- \rightleftharpoons \text{Cu}(\text{s})$	+0.34
$\text{Fe}^{3+}(\text{aq}) + \text{e}^- \rightarrow \text{Fe}^{2+}(\text{aq})$	+0.77
$\text{I}_2(\text{s}) + 2\text{e}^- \rightarrow 2\text{I}^-(\text{aq})$	+0.54
$\text{S}_2\text{O}_8^{2-}(\text{aq}) + 2\text{e}^- \rightarrow 2\text{SO}_4^{2-}(\text{aq})$	+2.01

An ion is deemed suitable to catalyse the reaction if it is able to be consumed and regenerated, i.e. the $E^\ominus$ value is in between that of 0.54V and +2.01V.

Cu^{2+} cannot oxidise iodide in the first place due to non-spontaneous reaction ($0.34 - 0.54 < 0$), so it cannot be consumed and is not a suitable catalyst.

Co^{2+} : reacts with peroxydisulfate to form SO_4^{2-} and Co^{3+} ($E^\ominus_{\text{cell}} = 2.01 - 1.82 > 0$) and Co^{3+} then reacts with iodide to give iodine and regenerates Co^{2+} ($E^\ominus_{\text{cell}} = 1.82 - 0.54 > 0$).

Fe^{2+} : reacts with peroxydisulfate to form SO_4^{2-} and Fe^{3+} ($E^\ominus_{\text{cell}} = 2.01 - 0.77 > 0$) and Fe^{3+} then reacts with iodide to give iodine and regenerates Fe^{2+} ($E^\ominus_{\text{cell}} = 0.77 - 0.54 > 0$).

Fe^{3+} : reacts with iodide to give iodine and Fe^{2+} ($E^\ominus_{\text{cell}} = 0.77 - 0.54 > 0$) and Fe^{2+} then reacts with peroxydisulfate to form SO_4^{2-} and regenerates Fe^{3+} ($E^\ominus_{\text{cell}} = 2.01 - 0.77 > 0$).

29 *Use of the Data Booklet is relevant to this question.*

Which one of the following is true about vanadium and its compounds?

- A** The maximum oxidation state of vanadium is found in VO^{2+} .
- B** V_2O_5 is used as a catalyst in the manufacture of ammonia.
- C** Zinc reduces $\text{VO}_2^+(\text{aq})$ to $\text{V}^{2+}(\text{aq})$.
- D** Vanadium is less dense than calcium.

Answer: **C**

Worked Solution:

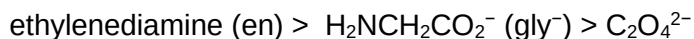
The maximum oxidation state of vanadium is found in VO_2^+

Finely divided Fe is used as a catalyst in the manufacture of ammonia

$E^\ominus_{\text{Zn}^{2+}/\text{Zn}} = -0.76\text{V}$; $E^\ominus_{\text{V}^{3+}/\text{V}^{2+}} = -0.26\text{V}$; $E^\ominus_{\text{V}^{2+}/\text{V}} = -1.20\text{V}$

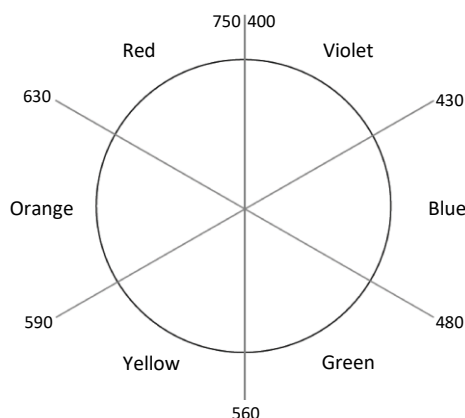
Vanadium is denser than calcium.

- 30 A partial spectrochemical series with ligands arranged in order of decreasing field strength is given below.



Stronger field ligands are known to give rise to a larger energy gap between the two sets of d-orbitals in a transition metal complex.

The figure below shows a colour wheel with approximate wavelength values (in nm) for different colour light. As wavelength decreases, the energy of the light increases.



An aqueous solution containing the $[\text{Co}(\text{gly})_3]$ complex is violet.

Which combination could be the colours of $[\text{Co}(\text{C}_2\text{O}_4)_3]^{3-}(\text{aq})$ and $[\text{Co}(\text{en})_3]^{3+}(\text{aq})$?

	$[\text{Co}(\text{C}_2\text{O}_4)_3]^{3-}(\text{aq})$	$[\text{Co}(\text{en})_3]^{3+}(\text{aq})$
A	green	yellow
B	yellow	orange
C	red	blue
D	blue	green

Answer: **A**

Worked Solution:

ethylenediamine (en) > $\text{H}_2\text{NCH}_2\text{CO}_2^- (\text{gly}^-) > \text{C}_2\text{O}_4^{2-}$

$[\text{Co}(\text{gly})_3]$ complex is violet, the complex absorbs yellow light (560–590 nm)

$[\text{Co}(\text{C}_2\text{O}_4)_3]^{3-}(\text{aq})$ a smaller energy gap between the two sets of d-orbitals in a transition metal complex, the complex absorbs a longer wavelength of light (590–700 nm), colour of complex should be **blue or green**

and $[\text{Co}(\text{en})_3]^{3+}(\text{aq})$, a larger energy gap between the two sets of d-orbitals in a transition metal complex, the complex absorbs a shorter wavelength light (400–590 nm), colour of complex should be **yellow, orange, red**